Andalusia Academy Course Catalogue: Artificial Intelligence

Synthetic demonstration data. Courses, fees, schedules, instructors and offers are illustrative and are not real Andalusia Academy information.

This catalogue lists the Artificial Intelligence courses of Andalusia Academy (Cairo campus (Nasr City)). Fees are in EGP. Online courses use: Live online classes on the academy's virtual classroom with recorded sessions.

Courses in this catalogue

- AI-101 Introduction to Artificial Intelligence and Python (Beginner, Online)
- AI-201 Machine Learning Practitioner (Intermediate, Hybrid (online sessions plus monthly on-campus labs in Cairo))
- AI-301 Building LLM Applications with RAG (Advanced, Online)

1. Introduction to Artificial Intelligence and Python

Course ID: AI-101
Department: Artificial Intelligence
Industry: Technology and AI
Level: Beginner
Delivery mode: Online
Language of instruction: Arabic and English
Duration: 8 weeks (48 hours)
Time commitment: 6 hours per week (two 3-hour sessions)
Fee (EGP): 6500
Payment options: Two payments of 3250 EGP
Certificate: Andalusia Academy certificate of completion
Accreditation: Not accredited (certificate of completion only)
Country: Egypt
Data status: Synthetic demonstration data
Updated: 2026-10-27

1.1 Course Description

A first course in AI for people with no programming background. Learners write Python from the first week and finish by building and explaining simple prediction models.
Target audience: Students, career changers and professionals from non-technical fields who want to start in AI.

1.2 Entry Requirements and Prerequisites

Entry requirements: No programming experience required. Basic computer skills and secondary-school mathematics.
Prerequisite courses: None

1.3 Learning Outcomes

- Write Python programs using variables, loops, functions and lists
- Load and explore a small dataset with pandas
- Explain what supervised learning is and when to use it
- Train and evaluate a simple classification model with scikit-learn

1.4 Topics Covered

- Python fundamentals
- Working with data in pandas
- What AI and machine learning are
- Your first classification model
- Evaluating models: accuracy and errors
- Mini project

1.5 Skills Taught

- Python
- pandas basics
- scikit-learn basics
- Problem framing

1.6 Career Outcomes

- Junior data or AI assistant roles

- Foundation for the Machine Learning Practitioner course

1.7 Schedule and Intakes

- Starts 2026-11-15: Sunday and Wednesday, 7:00-10:00 pm (Cairo time); 30 seats

- Starts 2027-01-10: Saturday and Tuesday, 6:00-9:00 pm (Cairo time); 30 seats

1.8 Instructors

- Dr. Laila Mansour, Lead AI Instructor (12 years of experience): Machine learning, deep learning, model evaluation

- Youssef Kamal, Data Science Instructor (7 years of experience): Python for data analysis, statistics, pandas

2. Machine Learning Practitioner

Course ID: AI-201
Department: Artificial Intelligence
Industry: Technology and AI
Level: Intermediate
Delivery mode: Hybrid (online sessions plus monthly on-campus labs in Cairo)
Language of instruction: English with Arabic support
Duration: 10 weeks (70 hours)
Time commitment: 7 hours per week (two 3.5-hour sessions)
Fee (EGP): 11500
Payment options: Three payments of about 3834 EGP
Certificate: Andalusia Academy certificate of completion
Accreditation: Not accredited (certificate of completion only)
Country: Egypt
Data status: Synthetic demonstration data
Updated: 2026-10-27

2.1 Course Description

Hands-on machine learning for learners who already code in Python. Covers the full workflow from data preparation to model selection, validation and communicating results.
Target audience: Developers, analysts and graduates who know Python and want practical machine-learning skills.

2.2 Entry Requirements and Prerequisites

Entry requirements: Comfortable with Python and basic statistics. AI-101 or equivalent experience.
Prerequisite courses: AI-101 Introduction to Artificial Intelligence and Python

2.3 Learning Outcomes

- Prepare data with feature engineering and pipelines
- Compare models with cross-validation
- Tune hyperparameters and avoid overfitting
- Present model results and limitations to non-technical stakeholders

2.4 Topics Covered

- Data preparation pipelines
- Regression and classification models
- Tree-based models and ensembles
- Cross-validation and tuning
- Imbalanced data
- Capstone project

2.5 Skills Taught

- scikit-learn
- Feature engineering
- Model evaluation
- Experiment tracking

2.6 Career Outcomes

- Junior machine learning engineer

- Data scientist (entry level)

2.7 Schedule and Intakes

- Starts 2026-12-01: Monday and Thursday, 7:00-10:30 pm (Cairo time); 25 seats

2.8 Instructors

- Dr. Laila Mansour, Lead AI Instructor (12 years of experience): Machine learning, deep learning, model evaluation

3. Building LLM Applications with RAG

Course ID: AI-301
Department: Artificial Intelligence
Industry: Technology and AI
Level: Advanced
Delivery mode: Online
Language of instruction: English
Duration: 8 weeks (56 hours)
Time commitment: 7 hours per week
Fee (EGP): 14000
Payment options: Two payments of 7000 EGP
Certificate: Andalusia Academy certificate of completion
Accreditation: Not accredited (certificate of completion only)
Country: Egypt
Data status: Synthetic demonstration data
Updated: 2026-10-27

3.1 Course Description

Design, build and evaluate applications on large language models: prompting, retrieval-augmented generation, tool use and safe deployment.
Target audience: Software engineers and ML practitioners who want to build production LLM features.

3.2 Entry Requirements and Prerequisites

Entry requirements: Solid Python, experience calling web APIs, and basic machine-learning knowledge (AI-201 or equivalent).
Prerequisite courses: AI-201 Machine Learning Practitioner

3.3 Learning Outcomes

- Design prompts and structured outputs for reliable behaviour
- Build a retrieval-augmented generation pipeline with a vector database
- Evaluate answer quality and grounding
- Plan deployment, cost and safety controls

3.4 Topics Covered

- How LLMs work
- Prompt design and structured output
- Embeddings and vector databases
- Retrieval-augmented generation
- Agents and tool use
- Evaluation, safety and deployment

3.5 Skills Taught

- Prompt engineering
- Embeddings and vector search
- RAG pipelines
- LLM evaluation

3.6 Career Outcomes

- AI engineer
- LLM application developer

3.7 Schedule and Intakes

- Starts 2027-01-17: Sunday and Wednesday, 7:00-10:30 pm (Cairo time); 20 seats

3.8 Instructors

- Omar Haddad, Senior AI Engineer and Instructor (9 years of experience): Large language models, retrieval-augmented generation, Python